

Project Executable Files

Date	1 October 2026
Team ID	SWTID-2026-2533
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	3 Marks

Project Executable Files

This document lists everything needed to run or deploy EduGenie: the submitted files, the folder structure, how to reach the live app, and how to run it locally.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile / desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Pending

Step 2: File / Folder Structure

```

EduGenie/
|-- main.py           FastAPI app and API endpoints
|-- gemini_client.py  shared Gemini helper (API key, model)
|-- explanation_module.py local LaMini-Flan-T5 explanations
|-- qna.py            question answering
|-- quiz_module.py    quiz generation and JSON validation
|-- summary_module.py summarization
|-- learning_path.py  learning recommendations
|-- templates/
|   |-- index.html    web page (HTML + JavaScript)
|-- static/
|   |-- style.css      styles
|-- requirements.txt   Python dependencies
|-- .env.example       environment template (copy to .env)
|-- .gitignore         keeps .env out of Git
|-- locustfile.py      load-test script
|-- README.md          setup and usage guide
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://edu-genie-mauve.vercel.app
Login Credentials (Demo)	Not required. EduGenie is open access with no sign-up or login.
Platform / Hosting Provider	Vercel (web hosting); Google AI Studio (Gemini API key)
Repository Link	https://github.com/bhavana-cse/EduGenie
Demo Video Link	https://drive.google.com/drive/folders/1TkpXJRq9Fv_TWc6seEcH5svR6fAOjVf?usp=drive_link

Step 4: Run Instructions

1. Install Python 3.10 or newer and open the project folder in Visual Studio Code.
2. Create and activate a virtual environment: `python -m venv .venv`, then `.venv\Scripts\activate` (Windows) or `source .venv/bin/activate` (macOS / Linux). If PowerShell blocks activation, run `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass` first.
3. Install the dependencies: `pip install -r requirements.txt`
4. Copy `.env.example` to `.env`, then set `GEMINI_API_KEY` (from Google AI Studio) and `GEMINI_MODEL`.
5. Start the server: `uvicorn main:app --reload`
6. Open `http://127.0.0.1:8000`, pick a task from the dropdown, type your text and press the button.
7. Optional: run the load test with `locust -f locustfile.py --host http://127.0.0.1:8000`.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	The first concept explanation downloads the local model (about 3 GB), so it is slow once.	Run one warm-up explanation beforehand; if the model cannot load, Gemini answers instead.
2	There is no login, saved history or progress tracking.	Planned as a future enhancement (see Scalability & Future Plan).
3	AI answers can occasionally be inaccurate or incomplete.	Learners should check important facts against textbooks or teachers.
4	Gemini features need internet and are limited by the API key's quota; Google also retires old model names.	Change <code>GEMINI_MODEL</code> in the environment settings; no code change is needed.